

Martin Luther University of Halle-Wittenberg
Department of Economics
Chair of Econometrics

Econometrics II
4. Panel data models

Christoph Wunder

Goals

- Use advantages of panel data
- Introduce within (fixed effects) estimator, random effects estimator
- Apply methods in R (packages: plm, lme4)

Outline

4 Panel data models

- 4.1 Introduction
- 4.2 Basic linear panel data models
 - 4.2.1 Fixed effects model
 - 4.2.2 Random effects model
 - 4.2.3 Choosing a model: Hausman test
 - 4.2.4 Example: cost function for airline production
- 4.3 Binary response models for panel data
 - 4.3.1 Conditional fixed effects model
 - 4.3.2 Random effects model
 - 4.3.3 Correlated random effects model
 - 4.3.4 Example: demand for health care

4.1 Introduction

4.2 Basic linear panel data models

4.2.1 Fixed effects model

4.2.2 Random effects model

4.2.3 Choosing a model: Hausman test

4.2.4 Example: cost function for airline production

4.3 Binary response models for panel data

4.3.1 Conditional fixed effects model

4.3.2 Random effects model

4.3.3 Correlated random effects model

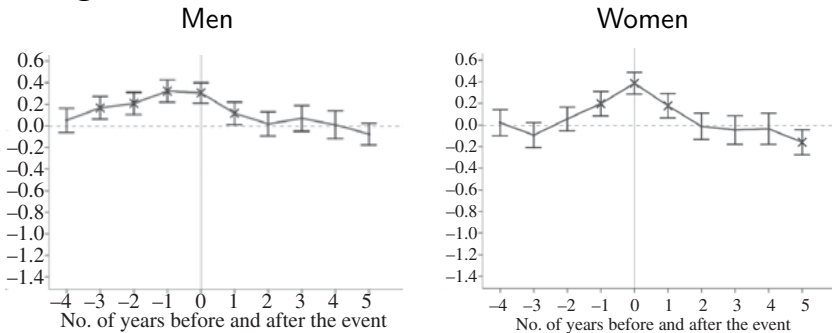
4.3.4 Example: demand for health care

4.1 Introduction

- **Panel data** (or **longitudinal data**) are data where **repeated observations for the same units** (individuals, households, firms, countries) are collected over a number of periods.
- In general, analyzing panel data is **more complicated** because it is usually no longer appropriate to assume that different observations are independent. (Note that we observe the same unit repeatedly.)
- **Advantages:** why should we use panel data?
 - Controlling for **individual heterogeneity** (e.g. unobserved time-invariant variables)
 - More informative data due to **more variability** (i.e. variability within and between units)
 - Analyze the **dynamics** of adjustment (e.g. transitions between labor market states, persistence in unemployment, welfare trap)
 - **Identify effects** that are not detectable in pure cross-section or pure time-series data (e.g. anchoring effects in life satisfaction)

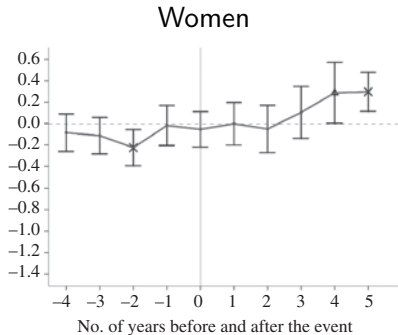
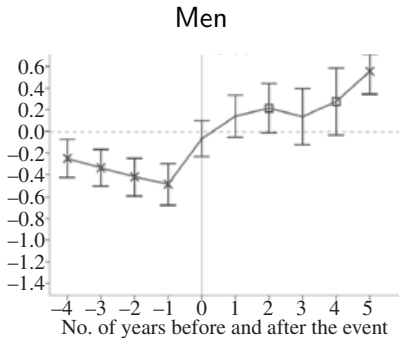
- **Example:** using panel data to analyze adaptation of life satisfaction to major life events

Marriage



Source: Clark et al. (2008), p. F234-F235

Men



- **Limitations**

- **Design and data collection problems** (e.g. frequency of interviewing, interview spacing)
- Distortions of **measurement errors** (e.g. due to recall, learning effects)
- **Selectivity** (e.g. non-response, attrition)
- **Short time-series dimension** (only short time span for each individual, incidental parameter problem in nonlinear models)
- A **panel data regression** has a double subscript on its variables:

$$y_{it} = \alpha_j + \mathbf{x}'_{it}\boldsymbol{\beta} + \epsilon_{it}. \quad (4.1)$$

- $\mathbf{x}_{it}$ is a K -dimensional vector of variables (**without** a constant).
- The effect of a change in $\mathbf{x}$ is the same for all units and all periods.
- The **individual-specific intercept**, α_i , allows that the average level for unit i may be different from that for unit j . It captures the effects of individual-specific, time-invariant variables.
- ϵ_{it} is assumed to be i.i.d. over individuals and time, with mean zero and variance σ_ϵ^2 .

- **Example:** using panel data to **address omitted variable bias**.
 - Omitted variable bias arises if a variable that is correlated with the included variables is excluded from the model.
 - Consider the estimation of a production function

$$y_{it} = \mu + \mathbf{x}'_{it}\boldsymbol{\beta} + m_i\beta_{K+1} + \epsilon_{it}, \quad (4.2)$$

where y_{it} denotes log output, x_{it} is a K -dimensional vector of log inputs, and m_i denotes the management quality of firm i (which is assumed to be constant over time).

- m_i is expected to be negatively correlated with the other inputs. Therefore, unless $\beta_{K+1} = 0$, deletion of m_i from the regression will lead to biased estimates of the other parameters.
- If panel data are available, this problem can be solved by introducing a firm specific effect $\alpha_i = \mu + m_i\beta_{K+1}$.

4.1 Introduction

4.2 Basic linear panel data models

4.2.1 Fixed effects model

4.2.2 Random effects model

4.2.3 Choosing a model: Hausman test

4.2.4 Example: cost function for airline production

4.3 Binary response models for panel data

4.3.1 Conditional fixed effects model

4.3.2 Random effects model

4.3.3 Correlated random effects model

4.3.4 Example: demand for health care

4.2 Basic linear panel data models

4.2.1 Fixed effects model

- The fixed effects model is simply a linear regression model in which the intercept terms vary over the individual units i :

$$y_{it} = \alpha_j + \mathbf{x}'_{it}\boldsymbol{\beta} + \epsilon_{it}, \quad \epsilon_{it} \sim \text{i.i.d.}(0, \sigma_\epsilon^2). \quad (4.3)$$

- We can write the model by including a set of N dummy variables:

$$y_{it} = \sum_{j=1}^N \alpha_j d_{ij} + \mathbf{x}'_{it} \boldsymbol{\beta} + \epsilon_{it}, \quad (4.4)$$

where $d_{ij} = 1$ if $i = j$, 0 otherwise.

- The parameters $\alpha_1, \dots, \alpha_N$ and β can be estimated by OLS. The estimator for β is called the **least squares dummy variable (LSDV) estimator**. However, the inclusion of many (additional) regressors may be unattractive.

- **Alternative approach:** the same estimator for β is obtained if the regression is performed in **deviations from individual means**. This eliminates the individual effects α_i .

$$(y_{it} - \bar{y}_j) = (\alpha_i - \alpha_j) + (\mathbf{x}_{it} - \bar{\mathbf{x}}_j)' \boldsymbol{\beta} + (\epsilon_{it} - \bar{\epsilon}_j), \quad (4.5)$$

where $\bar{y}_i = T^{-1} \sum_t y_{it}$. (Note that α_i drops out.)

- Equation 4.5 is called the **within transformation**. The OLS estimator for β obtained from this transformed model is called the **within estimator** or **fixed effects estimator**, denoted $\hat{\beta}_{FE}$. It is exactly **identical** to the **LSDV estimator**.
- If we assume $E[(\mathbf{x}_{it} - \bar{\mathbf{x}}_i)\epsilon_{it}] = \mathbf{0}$, then $\hat{\beta}_{FE}$ is **consistent** (for $N \rightarrow \infty$ and fixed T) (strict exogeneity assumptions on the explanatory variables, see Wooldridge 2010, Ch. 10.2.2).
- This assumption implies that explanatory variables in each time period are uncorrelated with the idiosyncratic error in each time period:
 $E[\mathbf{x}_{is}\epsilon_{it}] = 0$ for all $s, t = 1, \dots, T$.
- The assumption does allow arbitrary correlation between α_i and $\mathbf{x}_{is}$.

- With explanation

- $$\hat{\alpha}_i = \bar{y}_i - \bar{\mathbf{x}}_i' \hat{\boldsymbol{\beta}}_{FE}. \quad (4.6)$$

- $$Var(\hat{\beta}_{FE}) = \sigma_{\epsilon}^2 \left(\sum_{i=1}^N \sum_{t=1}^T (\mathbf{x}_{it} - \bar{\mathbf{x}}_i)(\mathbf{x}_{it} - \bar{\mathbf{x}}_i)' \right)^{-1}. \quad (4.7)$$

- $$\hat{\sigma}_\epsilon^2 = \frac{1}{NT - N - K} \sum_{i=1}^N \sum_{t=1}^T \left((y_{it} - \bar{y}_i) - (\mathbf{x}_{it} - \bar{\mathbf{x}}_i)' \hat{\boldsymbol{\beta}}_{FE} \right)^2. \quad (4.8)$$

- 11 / 46

- When interpreting $\hat{\beta}_{FE}$, it may be important to realize that the parameters are **identified only through the within dimension** of the data.
- That is, the fixed effects model explains to what extent y_{it} differs from $\bar{y}_i$ and does not explain why $\bar{y}_i$ is different from $\bar{y}_j$.
- The parametric assumptions about β impose that a change in x has the same (ceteris paribus) effect whether it is a change from one period to the other or a change from one individual to the other.

4.2.2 Random effects model

- The **random effects model** assumes that the α_i s are random, which reduces the number of parameters considerably.

$$y_{it} = \mu + \mathbf{x}'_{it}\boldsymbol{\beta} + \alpha_i + \epsilon_{it}, \quad \epsilon_{it} \sim \text{i.i.d.}(0, \sigma_\epsilon^2), \quad \alpha_i \sim \text{i.i.d.}(0, \sigma_\alpha^2). \quad (4.9)$$

- **Assumptions**
 - The **individual specific error term component**, α_i , does not vary over time.
 - The **idiosyncratic (i.e. remainder) error term component**, ϵ_{it} , is uncorrelated over time. All correlation of the error terms over time is attributed to the individual effects α_i .
 - α_i and ϵ_{it} are mutually independent and independent of $\mathbf{x}_{js}$ ($\forall j, s$).
- The assumptions imply that the OLS estimator for μ and $\boldsymbol{\beta}$ in equation 4.9 is unbiased and consistent. However, OLS standard errors are incorrect. An efficient (F)GLS estimator can be obtained by exploiting the structure of the error covariance matrix.

Derivation of the **covariance matrix** of the random effects model

- For one individual, we have

$$E(\epsilon_{it}) = E(\alpha_i) = 0 \quad (4.10)$$

$$E(\epsilon_{it}\epsilon_{is}) = 0 \quad \forall t \neq s \quad (4.11)$$

$$E(\alpha_i\epsilon_{it}) = 0 \quad \forall t \quad (4.12)$$

$$\begin{aligned} E((\alpha_i + \epsilon_{it})(\alpha_i + \epsilon_{is})) &= E(\alpha_i^2) + E(\alpha_i\epsilon_{is}) + E(\epsilon_{it}\alpha_i) + E(\epsilon_{it}\epsilon_{is}) \\ &= \begin{cases} \sigma_\alpha^2 + \sigma_\epsilon^2, & \text{for } t = s; \\ \sigma_\alpha^2, & \text{for } t \neq s. \end{cases} \end{aligned}$$

- Defining the composite error term as $\eta_{it} = \alpha_i + \epsilon_{it}$, we have

$$E(\eta\eta') = \sigma_\alpha^2(\mathbf{I}_N \otimes \mathbf{J}_T) + \sigma_\epsilon^2(\mathbf{I}_N \otimes \mathbf{I}_T), \quad (4.13)$$

where $\mathbf{I}_T$ is the T -dimensional identity matrix, $\mathbf{I}_N$ is the N -dimensional identity matrix, and $\mathbf{J}_T = \mathbf{i}_T \mathbf{i}_T'$ (with $\mathbf{i}_T = (1, 1, \dots, 1)'$).

- **Example:** covariance matrix for $n = 2$ and $T = 3$:

$$E(\eta\eta') = \begin{pmatrix} \sigma_\epsilon^2 + \sigma_\alpha^2 & \sigma_\alpha^2 & \sigma_\alpha^2 & 0 & 0 & 0 \\ \sigma_\alpha^2 & \sigma_\epsilon^2 + \sigma_\alpha^2 & \sigma_\alpha^2 & 0 & 0 & 0 \\ \sigma_\alpha^2 & \sigma_\alpha^2 & \sigma_\epsilon^2 + \sigma_\alpha^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma_\epsilon^2 + \sigma_\alpha^2 & \sigma_\alpha^2 & \sigma_\alpha^2 \\ 0 & 0 & 0 & \sigma_\alpha^2 & \sigma_\epsilon^2 + \sigma_\alpha^2 & \sigma_\alpha^2 \\ 0 & 0 & 0 & \sigma_\alpha^2 & \sigma_\alpha^2 & \sigma_\epsilon^2 + \sigma_\alpha^2 \end{pmatrix} \quad (4.14)$$

- ① Homoskedasticity: $E(\eta_{it}, \eta_{js}) = \sigma_\epsilon^2 + \sigma_\alpha^2$ for $i = j; t = s$
- ② Particular form of autocorrelation:

$$E(\eta_{it}, \eta_{js}) = \begin{cases} \sigma_\alpha^2, & i = j; t \neq s; \\ 0, & i \neq j. \end{cases} \quad (4.15)$$

Estimating the random effects model using GLS

- The GLS transformation is

$$\left(y_{it} - \bar{y}_i + \psi^{\frac{1}{2}} \bar{y}_i\right) = \left(\mathbf{x}_{it} - \bar{\mathbf{x}}_i + \psi^{\frac{1}{2}} \bar{\mathbf{x}}_i\right)' \boldsymbol{\beta} + \eta_{it}^*, \quad (4.16)$$

where

$$\psi = \frac{\sigma_{\epsilon}^2}{\sigma_{\epsilon}^2 + T\sigma_{\alpha}^2}$$

- For $\psi \approx 1$: GLS estimator $\Rightarrow$ pooled OLS estimator
 - $\sigma_{\epsilon}^2 \gg \sigma_{\alpha}^2$
 - $\sigma_{\alpha}^2 = 0$
- For $\psi \approx 0$: GLS estimator $\Rightarrow$ within estimator
 - $\sigma_{\epsilon}^2 \rightarrow 0$
 - $T \rightarrow \infty$
- A FGLS estimator can be obtained by estimating the variance components σ_{ϵ}^2 and σ_{α}^2 from the within regression and the between regression, respectively (e.g. Swamy and Arora 1972).
- Pooled OLS (with cluster-robust s.e.), ML estimation, or Bayesian methods are alternatives.

4.2.3 Choosing a model: Hausman test

- The **fixed effects model** estimates many parameters. The loss of degrees of freedom can be avoided if the α_i s can be assumed to be random.
- However, the **random effects model** assumes that the individual-specific error term component, α_i , and the explanatory variables, $\mathbf{x}_{it}$, are uncorrelated.
- The **null hypothesis** of the **Hausman test** states that this (critical) assumption of the random effects model is satisfied.
- If α_i and $\mathbf{x}_{it}$ are uncorrelated, then both the fixed effects estimator and the random effects estimator are consistent, and the random effects estimator is efficient.
- If α_i and $\mathbf{x}_{it}$ are correlated, then the random effects estimator is inconsistent.

- The **Hausman test statistic** is

$$\xi_H = (\hat{\beta}_{FE} - \hat{\beta}_{RE})' [\widehat{Var(\hat{\beta}_{FE})} - \widehat{Var(\hat{\beta}_{RE})}]^{-1} (\hat{\beta}_{FE} - \hat{\beta}_{RE}), \quad (4.17)$$

where $\widehat{Var}(\cdot)$ are estimates of the true covariance matrices.

- Under the null hypothesis, which implicitly says that $\text{plim}(\hat{\beta}_{FE} - \hat{\beta}_{RE}) = \mathbf{0}$, the statistic ξ_H has an asymptotic χ^2 distribution with K degrees of freedom, where K is the number of elements in β .
- The Hausman test thus tests whether the fixed effects and random effects estimator are significantly different.
- A practical problem in computing the test statistic is that the covariance matrix in square brackets may not be positive definite in finite samples, such that its inverse cannot be computed. As an alternative, it is possible to test for a subset of the elements in β .

4.2.4 Example: cost function for airline production

Source: Greene (2003), Ch. 13

- **Research objective:** we estimate a regression model for the total cost of airline production.

$$\ln c_{it} = \beta_0 + \beta_1 \ln q_{it} + \beta_2 \ln pf_{it} + \beta_3 lf_{it} + \alpha_i + \epsilon_{it} \quad (4.18)$$

- **Data:**

I = Airline

T = Year

Q = Output, in revenue passenger miles, index number

C = Total cost, in \$1000

PF = Fuel price

LF = Load factor, the average capacity utilization of the fleet

- Load the data in R.

```
d <- read.csv("data/TableF6-1.csv", header=TRUE, sep=",")
str(d, width=60, strict.width=c("cut"))

## 'data.frame': 90 obs. of 6 variables:
## $ I : int 1 1 1 1 1 1 1 1 1 1 ...
## $ T : int 1 2 3 4 5 6 7 8 9 10 ...
## $ C : int 1140640 1215690 1309570 1511530 1676730 182374..
## $ Q : num 0.953 0.987 1.092 1.176 1.16 ...
## $ PF: int 106650 110307 110574 121974 196606 265609 2634..
## $ LF: num 0.534 0.532 0.548 0.541 0.591 ...
```

- We have a **balanced panel**.

```
library(plm)
d <- pdata.frame(d, index=c("I", "T"))
pdim(d)

## Balanced Panel: n = 6, T = 15, N = 90

d[c(1:3,15:18), ]

##           I   T           C           Q           PF           LF
## 1-1      1   1  1140640  0.952757  106650  0.534487
## 1-2      1   2  1215690  0.986757  110307  0.532328
## 1-3      1   3  1309570  1.091980  110574  0.547736
## 1-15     1  15  4748320  1.936460  823411  0.625603
## 2-1      2   1   569292  0.520635  103795  0.490851
## 2-2      2   2   640614  0.534627  111477  0.473449
## 2-3      2   3   777655  0.655192  118664  0.503013
```

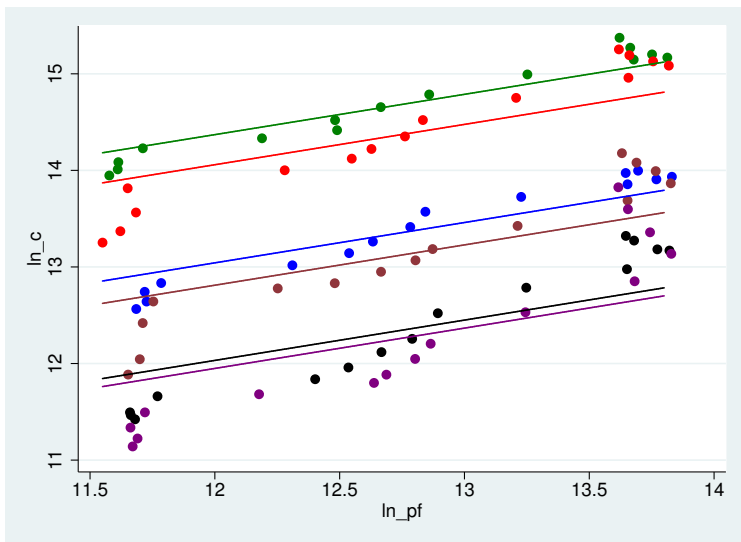
Basic linear panel data models

- LSDV estimates:

```
lsdv <- lm(log(C) ~ -1 + log(Q) + log(PF) + LF + as.factor(I),
           data = d)
round(summary(lsdv)$coefficients, 4)
```

##		Estimate	Std. Error	t value	Pr(> t)
##	log(Q)	0.9193	0.0299	30.7555	0
##	log(PF)	0.4175	0.0152	27.4682	0
##	LF	-1.0704	0.2017	-5.3071	0
##	as.factor(I) 1	9.7059	0.1931	50.2576	0
##	as.factor(I) 2	9.6647	0.1990	48.5708	0
##	as.factor(I) 3	9.4970	0.2250	42.2168	0
##	as.factor(I) 4	9.8905	0.2418	40.9099	0
##	as.factor(I) 5	9.7300	0.2609	37.2880	0
##	as.factor(I) 6	9.7930	0.2637	37.1423	0

- Graphical representation of LSDV estimates:



- Within estimator:

```
fe <- plm(log(C) ~ log(Q) + log(PF) + LF,
          data = d, model = "within")
round(summary(fe)$coefficients, 4)

##              Estimate Std. Error t-value Pr(>|t|)
## log(Q)          0.9193     0.0299 30.7555      0
## log(PF)          0.4175     0.0152 27.4682      0
## LF              -1.0704     0.2017 -5.3071      0
```

- Random effects estimator (FGLS):

```
re <- plm(log(C) ~ 1 + log(Q) + log(PF) + LF,
          data = d, model = "random")
round(summary(re)$coefficients, 4)

##              Estimate Std. Error z-value Pr(>|z|)
## (Intercept)    9.6279     0.2102 45.8114      0
## log(Q)          0.9067     0.0256 35.3827      0
## log(PF)          0.4228     0.0140 30.1451      0
## LF              -1.0645     0.2001 -5.3206      0
```

- Estimation results:

	OLS		LSDV		FEM		REM	
	Beta	S.E.	Beta	S.E.	Beta	S.E.	Beta	S.E.
(Intercept)	9.52	0.23	-	-	-	-	9.63	0.21
log(Q)	0.88	0.01	0.92	0.03	0.92	0.03	0.91	0.03
log(PF)	0.45	0.02	0.42	0.02	0.42	0.02	0.42	0.01
LF	-1.63	0.35	-1.07	0.2	-1.07	0.2	-1.06	0.20

- **Hausman test**

```

phtest(fe, re)

##
##   Hausman Test
##
## data:  log(C) ~ log(Q) + log(PF) + LF
## chisq = 2.1247, df = 3, p-value = 0.5469
## alternative hypothesis: one model is inconsistent

```

We do not reject the the null hypothesis of the Hausman test. This suggests that the individual effects and the explanatory variables are uncorrelated. Hence, the random effects model appears to be an appropriate choice.

- Estimation of the random effects model using maximum likelihood:

```
library(lme4)
lme <- lmer(log(C) ~ 1 + log(Q) + log(PF) + LF + (1|I),
            data = d)
summary(lme)$coefficient
```

##		Estimate	Std. Error	t value
##	(Intercept)	9.6322117	0.21155881	45.529713
##	log(Q)	0.9073167	0.02580895	35.155125
##	log(PF)	0.4225032	0.01405981	30.050433
##	LF	-1.0645720	0.19977606	-5.328827

- Estimation of the random effects model using MCMC:

```
library(brms)
bayes <- brm(log(C) ~ 1 + log(Q) + log(PF) + LF + (1|I),
             data = d, family = gaussian(),
             control = list(adapt_delta = 0.99))
```

```
fixef(bayes)
```

##	Estimate	Est.Error	Q2.5	Q97.5
## Intercept	9.6474624	0.22925313	9.1936271	10.0988153
## logQ	0.9099226	0.02719059	0.8571206	0.9640340
## logPF	0.4214813	0.01447309	0.3939238	0.4500854
## LF	-1.0675287	0.19801035	-1.4483741	-0.6742285

Outline

4 Panel data models

4.1 Introduction

4.2 Basic linear panel data models

4.2.1 Fixed effects model

4.2.2 Random effects model

4.2.3 Choosing a model: Hausman test

4.2.4 Example: cost function for airline production

4.3 Binary response models for panel data

4.3.1 Conditional fixed effects model

4.3.2 Random effects model

4.3.3 Correlated random effects model

4.3.4 Example: demand for health care

4.3 Binary response models for panel data

Greene (2011), Ch. 17.4

- Standard model for panel data:

$$y_{it}^* = \mathbf{x}_{it}'\boldsymbol{\beta} + \alpha_i + \epsilon_{it}, \quad i = 1, \dots, N, \quad t \in T_i \subset \{1, \dots, T\}, \quad (4.19)$$

where α_i represents unobserved, individual specific heterogeneity.

- If we ignore heterogeneity, the ML estimator will be inconsistent (even if α_i and $\mathbf{x}_{it}$ are uncorrelated) because the likelihood function is misspecified.
- Solutions:
 - Fixed effects models
 - Random effects models

4.3.1 Conditional fixed effects model

- Incidental parameters problem:
 - Each α_i needs to be estimated from T_i observations.
 - ML estimator requires large T_i so that the α_i s are consistently estimated.
 - However, in typical panel data setting, T_i is small while N is large.
 - The ML estimators of the α_i s and of β are inconsistent even if the model is properly specified.
- Solution: conditional fixed effects estimator
- Chamberlain's conditional likelihood function

$$L^c = \prod_{i=1}^N P \left(y_{i1}, \dots, y_{iT_i} \left| \sum_{t=1}^{T_i} y_{it} \right. \right) \quad (4.20)$$

does not depend on the incidental parameters, α_i .

- Disadvantages:
 - The estimator uses only data on individuals with variation over time. Therefore, the conditional likelihood is typically based on a restricted data set.
 - Approach cannot be applied with probit model.
 - Under homogeneity ($\alpha_i = \alpha, \forall i$), Chamberlain's conditional ML estimator (CMLE) is inefficient (but still consistent). Hausman specification test can be used to test the null hypothesis of homogeneity.
 - H_0 (homogeneity): CMLE and usual MLE are consistent, but CMLE is inefficient, as it fails to use the information that $\alpha_i = \alpha$ and it may not use all the data.
 - H_1 (heterogeneity): usual MLE is inconsistent whereas CMLE is consistent and efficient.
 - Without estimates for the α_i s, we cannot compute predicted probabilities or marginal effects. Only inferences about ratios of coefficient can be made.

4.3.2 Random effects model

- The random effects model is based on the individual likelihood contributions:

$$L_i = \int_{-\infty}^{+\infty} \prod_{t=1}^{T_i} P(y_{it} | \mathbf{x}'_{it}\boldsymbol{\beta} + \alpha_i) f(\alpha_i) d\alpha_i \quad (4.21)$$

- The probability may be probit or logit.
- ML: Gaussian quadrature is used to integrate out the random effect.
- MCMC is a good alternative for multilevel model.
- Disadvantages: strong assumptions
 - Model assumes normality for α_i .
 - Model assumes equal correlation across periods.
 - Model assumes that α_i and $\mathbf{x}_{it}$ are independent. However, Mundlak's approach may be used.

4.3.3 Correlated random effects model

Mundlak (1978)

- Standard model for panel data is

$$y_{it}^* = \mathbf{x}_{it}'\boldsymbol{\beta} + \alpha_i + \epsilon_{it}, \quad (4.22)$$

$$P(y_{it}|\mathbf{x}_{it}) = F(\alpha_i + \mathbf{x}_{it}'\boldsymbol{\beta}) \quad (4.23)$$

- We take account of the potential correlation between $\mathbf{x}_{it}$ and α_i by introducing an auxiliary regression:

$$\alpha_i = \alpha + \bar{\mathbf{x}}_i'\boldsymbol{\delta} + u_i, \quad (4.24)$$

where $\bar{\mathbf{x}}_i$ are individual-specific averages of the time varying covariates and u_i is an individual-specific error.

- The reduced form model is a random effects model:

$$P(y_{it}|\mathbf{x}_{it}) = F(\alpha + \bar{\mathbf{x}}_i'\boldsymbol{\delta} + \mathbf{x}_{it}'\boldsymbol{\beta} + u_i). \quad (4.25)$$

4.3.4 Example: demand for health care

Data source: Riphahn et al. (2003)

- Riphahn et al. (2003) analyze health care utilization using data from the Socio-Economic Panel.
- Outcome variable `docvis01` = 1 if the individual visited the doctor in the last three months; = 0 otherwise.

Variable	Mean	Std.Dev.	Min.	Max.	n
year	1987.82	3.17	1984	1994.00	27326
docvis01	0.63	0.48	0	1.00	27326
age	43.53	11.33	25	64.00	27326
hhninc	3.52	1.77	0	30.67	27326
hhkids	0.40	0.49	0	1.00	27326
educ	11.32	2.32	7	18.00	27326
married	0.76	0.43	0	1.00	27326

37 / 46

```
d <- d %>% group_by(id) %>%
  mutate( m_age = mean(age), m_hhninc = mean(hhninc),
          m_hhkids = mean(hhkids), m_educ = mean(educ),
          m_married = mean(as.numeric(married)-1) )
d[c(1:7), c("id", "year", "age", "m_age")]
```

```
## # A tibble: 7 x 4
## # Groups:   id [2]
##   id     year   age m_age
##   <fct> <fct> <int> <dbl>
## 1 1     1984     54    55
## 2 1     1985     55    55
## 3 1     1986     56    55
## 4 2     1984     44   45.8
## 5 2     1985     45   45.8
## 6 2     1986     46   45.8
## 7 2     1988     48   45.8
```


Fuller data models

CORRELATED RANDOM EFFECTS

```
data = d, family = binomial(link = "logit") )
```

- Estimation results:

	Pooled		CMLE		REM		CREM	
	Beta	S.E.	Beta	S.E.	Beta	S.E.	Beta	S.E.
Intercept	0.25	0.09	NA	NA	0.10	0.16	0.64	0.17
Age	0.02	0.00	0.08	1.09	0.03	0.00	0.08	0.01
Income	-0.02	0.01	-0.01	0.99	0.00	0.01	-0.01	0.02
Kids	-0.23	0.03	-0.08	0.93	-0.27	0.05	-0.07	0.07
Educ.	-0.05	0.01	-0.09	0.91	-0.06	0.01	-0.10	0.06
Married	0.09	0.03	-0.05	0.95	0.03	0.05	-0.05	0.09

- Results vary substantially across estimators; difficult to compare.
- Fixed effects estimators drop any individuals that have no variation over time in the dependent variable (approx. 40% loss).

- Likelihood ratio test for random effects:

```
library(lmtest)
lrtest(m1, m3)

## Likelihood ratio test
##
## Model 1: docvis01 ~ age + hhninc + hhkids + educ + married
## Model 2: docvis01 ~ age + hhninc + hhkids + educ + married + (1 | id)
##      #Df LogLik Df   Chisq Pr(>Chisq)
## 1      6 -17673
## 2      7 -16400  1 2546.1  < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

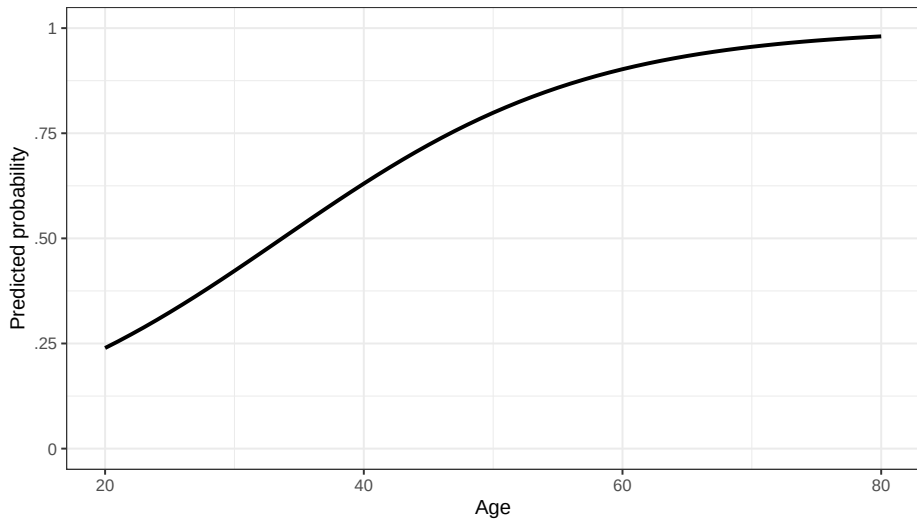
- The likelihood ratio test of the random effects logit model against the pooled logit rejects the restricted pooled logit ($H_0 : \sigma_\alpha = 0$).

- Average marginal effects:

```
library(margins)
ame.m1 <- margins( m1, data = d )
ame.m3 <- margins( m3, data = d )
ame.m4 <- margins( m4, data = d,
                   variables = c("age", "hhninc", "hhkids",
                                "educ", "married") )
```

	Pooled	REM	CREM
age	0.0047	0.0060	0.0154
educ	-0.0104	-0.0105	-0.0174
hhkids	-0.0522	-0.0487	-0.0133
hhninc	-0.0042	-0.0004	-0.0009
married1	0.0195	0.0060	-0.0082

```
type = "response", allow.new.levels = TRUE)
```



Key terms

- balanced panel
- fixed effects estimator
- Hausman test
- idiosyncratic error
- individual heterogeneity
- individual-specific intercept
- longitudinal data
- LSDV estimator
- panel data
- random effects model
- unbalanced panel
- within estimator
- within transformation

References I

This chapter is based on Verbeek (2012), Chapter 10.

Further readings: Greene (2011), Chapter 13

Clark, A. E., Diener, E., Georgellis, Y., and Lucas, R. E. (2008). Lags and leads in life satisfaction: A test of the baseline hypothesis. *Economic Journal*, 118(529):F222–F243.

Greene, W. H. (2003). *Econometric Analysis*. Prentice Hall International, Upper Saddle River, New Jersey.

Greene, W. H. (2011). *Econometric Analysis*. Prentice Hall International, Upper Saddle River, New Jersey.

Mundlak, Y. (1978). On the pooling of time series and cross section data. *Econometrica*, 46(1):69–85.

Riphahn, R., Wambach, A., and Million, A. (2003). Incentive effects in the demand for health care: A bivariate panel count data estimation. *Journal of Applied Econometrics*, 18(4):387–405.

Swamy, P. A. V. B. and Arora, S. S. (1972). The exact finite sample properties of the estimators of coefficients in the error components regression models. *Econometrica*, 40(2):261–275.

Verbeek, M. (2012). *A Guide to Modern Econometrics*. John Wiley & Sons, Chichester.

Wooldridge, J. M. (2010). *Econometric Analysis of Cross Section and Panel Data*. MIT Press, Cambridge, Massachusetts.